

Minjeong Kim

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Education

Ph.D. in Statistics, Ewha Womans University	Sep 2022 – Aug 2025
MBA in AI Big Data, aSSIST	Mar 2020 – Feb 2022
M.S. in Statistics, Ewha Womans University	Sep 2013 – Aug 2015
B.S. in Statistics (Minor in Law), Ewha Womans University	Mar 2008 – Aug 2013

Professional Experience

Data Scientist, Lunit Inc. – Cancer Screening AI Research Dept. Mar 2021 – Present

- Designed and led development of `research-cs-evaluation`, an internal library for lesion-level performance metrics (ROC, FROC, AFROC, bootstrapping, DeLong test, etc.), now adopted across all cancer screening AI pipelines.
- Developed a multi-anchor threshold shifting strategy to guarantee predefined sensitivity levels at product thresholds; collaborated with MA, PM, and Marketing teams to align statistical rigor with clinical usability and regulatory requirements.
- Performed model score calibration (e.g., Platt scaling, isotonic regression) to align AI-predicted probabilities with observed cancer prevalence, supporting reliable threshold selection and clinical interpretability.
- Built a model consistency evaluation framework for AI releases, defining acceptance criteria that capture user-impacting prediction changes to ensure clinical interpretability and stability.
- Led statistical validation of MMG and DBT AI models on multi-institutional datasets (KBSMC and external hospitals), supporting peer-reviewed publications and clinical study planning.
- Researched machine learning integration of AI-derived mammography and density scores (4-view, 8 features) to assess potential improvements in breast cancer detection.
- Conducted FDA submission-oriented analyses for DBT models, including large-scale performance evaluation and development of automated reporting code to generate regulatory-ready result summaries.
- Co-authored 6+ peer-reviewed publications in international journals and conferences, contributing expertise in study design, statistical validation, and AI evaluation methodology.

Biostatistician / Data Manager, Medytox Inc. May 2017 – Feb 2021

- Conducted statistical analysis for 10+ Phase I/III clinical trials (e.g., CORTOX, MT921), including SAP design, efficacy/safety analysis, and CDISC-standard reporting.
- Developed validated SAS programs for endpoint evaluation and regulatory submissions.
- Prepared ADaM/SDTM datasets and wrote statistical sections of integrated reports.
- Led data review and blind meeting preparations in collaboration with clinical, QA, and RA teams.
- Designed CRFs and DVPs, executed UAT, and supported EDC system deployment as a dual-role data manager.

Biostatistician, Catholic Medical Center Aug 2016 – May 2017

- Provided statistical consulting for investigator-initiated research across multiple hospitals within the Catholic Medical Center network.
- Supported clinicians in study design, sample size calculation, and statistical interpretation for diverse clinical specialties.
- Conducted survival analysis (Cox), PSM, GLMM, and AUC comparison (DeLong) using R.
- Authored IRB proposal sections and contributed to several clinical publications.

Publications

Accelerated increase in ferritin levels during menopausal transition as a marker of metabolic health Aug 2025

Minjeong Kim, Yoosoo Chang, Yoosun Cho, Min-Jung Kwon, Hee-Kyung Joh, Ga-Young Lim, Ria Kwon, Jiin Ahn, Jungeun Park, Kye-Hyun Kim, Jae Keun Yoo, Seungho Ryu

Scientific Reports, Aug 2025
DOI: 10.1038/s41598-025-14295-3

Hearing changes and trajectories during the menopausal transition and their association with metabolic factors

Oct 2025

Yoonyoung Jang, Yoosoo Chang, Jieun Lee, Byungtae Seo, Yoosun Cho, **Minjeong Kim**, Jae Ho Park, Jeonggyu Kang, Ria Kwon, Ga-young Lim, Kye-Hyun Kim, Hoon Kim, Yun Soo Hong, Jihwan Park, Di Zhao, Juhee Cho, Eliseo Guallar, Seungho Ryu
Maturitas, Oct 2025
DOI: 10.1016/j.maturitas.2025.108686

From Pixel Scores to Clinical Impacts: The Implicit Choices in FROC Metric Design and Their Consequences

Feb 2025

Minjeong Kim, Hesham Dar, Sanguk Park, Thijs Kooi
AMAI 2024, Oct 2024 (Proceedings published Feb 2025)
DOI: 10.1007/978-3-031-82007-6_2

Concordant and discordant breast density patterns by different approaches for assessing breast density and breast cancer risk

Nov 2024

Yoosun Cho, Eun Kyung Park, Yoosoo Chang, Mi-Ri Kwon, Eun Young Kim, **Minjeong Kim**, Boyoung Park, Sanghyup Lee, Han Eol Jeong, Ki Hwan Kim, Tae Soo Kim, Hyeonsoo Lee, Ria Kwon, Ga-Young Lim, JunHyeok Choi, Shin Ho Kook, Seungho Ryu
Breast Cancer Research and Treatment, Nov 2024
DOI: 10.1007/s10549-024-07541-1

Artificial Intelligence-Powered Imaging Biomarker Based on Mammography for Breast Cancer Risk Prediction

Jun 2024

Eun Kyung Park, Hyeonsoo Lee, **Minjeong Kim**, Taesoo Kim, Junha Kim, Ki Hwan Kim, Thijs Kooi, Yoosoo Chang, Seungho Ryu
Diagnostics, Jun 2024
DOI: 10.3390/diagnostics14121212

Pseudonymization's Effect on Data Quality: A Study Under Personal Information Protection Act

Jun 2024

Minjeong Kim, Jae Keun Yoo
The Korean Journal of Applied Statistics, Jun 2024
DOI: 10.5351/KJAS.2024.37.3.381

Screening mammography performance according to breast density: a comparison between radiologists versus standalone intelligence detection

Apr 2024

Mi-Ri Kwon, Yoosoo Chang, Seungho Ryu, Soo-Youn Ham, **Minjeong Kim**, et al.
Breast Cancer Research, Apr 2024
DOI: 10.1186/s13058-024-01821-w

Enhancing Breast Cancer Risk Prediction by Incorporating Prior Images

Oct 2023

Hyeonsoo Lee, Junha Kim, Eun Kyung Park, **Minjeong Kim**, Taesoo Kim, Thijs Kooi
MICCAI 2023 Proceedings, Oct 2023
DOI: 10.1007/978-3-031-43904-9_38

Validation study of machine-learning chest radiograph software in primary and emergency medicine

Sep 2022

E.J.R van Beek, J.S Ahn, **Minjeong Kim**, J.T Murchison
Clinical Radiology, Sep 2022
DOI: 10.1016/j.crad.2022.08.129

Academic Service

Manuscript Reviewer, *npj Aging* (Springer Nature)
Reviewed 1 manuscript in 2025

2025

Skills

Programming: Python, R, SQL

Tools: Git, Docker

Languages: English (fluent), Korean (native)